Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**53**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T1B**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-A2-A0EW-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.5** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

76th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

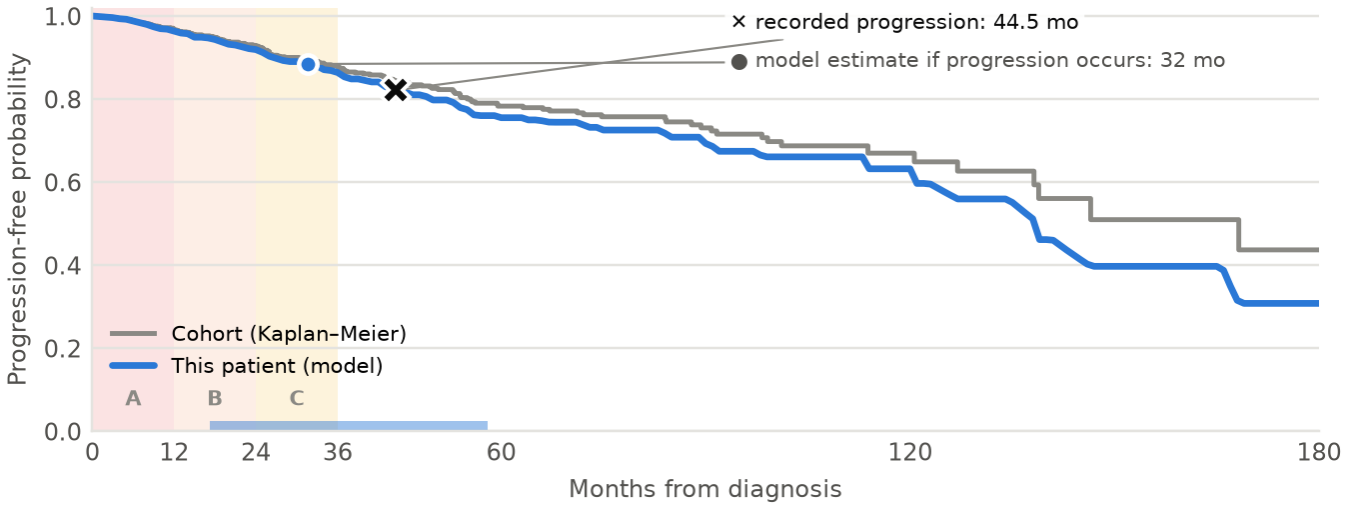
≈ 32 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

44.5 months

Group D · Late progression
✗ recorded window differs · |error| 13 mo



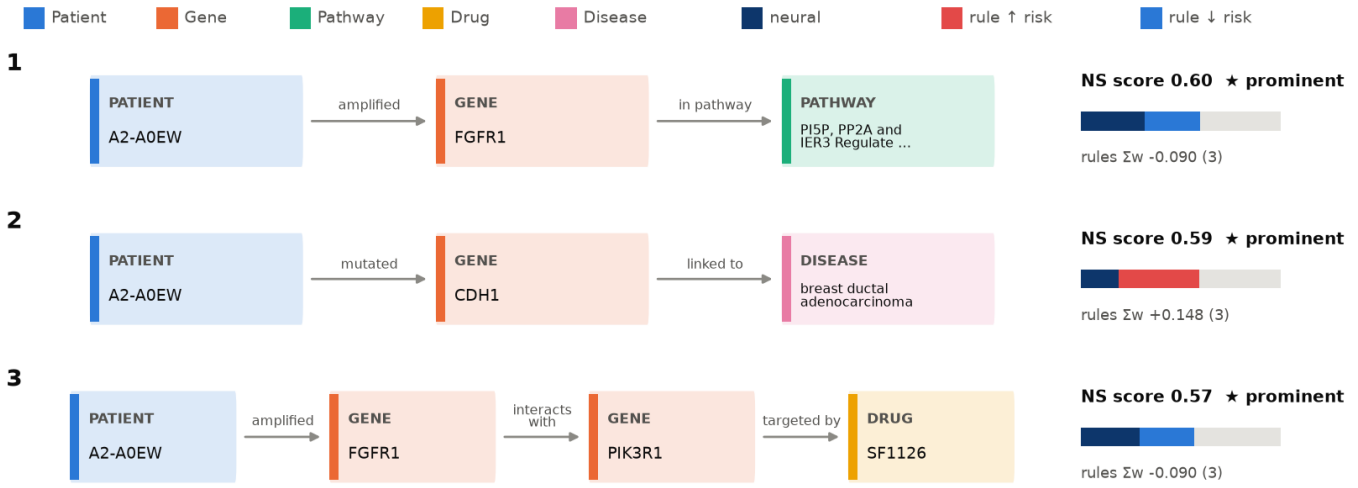
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 53-year-old patient has a LumA tumour, AJCC stage III (T1, N3, M0). The fused neuro-symbolic model places them in the **76th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T1)* ∧ *Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.125 per SD). The highest-scoring knowledge-graph path runs from **FGFR1** to **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**; the learned formulas it instantiates point to slower progression (Σw -0.090). Progression was recorded at 44.5 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

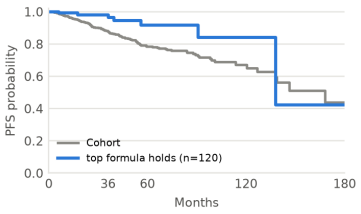


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A0EW** amplified → **FGFR1** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	54	↓ **slower progression** · Σw -0.090 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.73, n = 536 carriers	**0.60**
2 ★	**TCGA-A2-A0EW** mutated → **CDH1** linked to → **breast ductal adenocarcinoma**	32	↑ **faster progression** · Σw +0.148 over 3 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.29, n = 521 carriers	**0.59**
3 ★	**TCGA-A2-A0EW** amplified → **FGFR1** interacts with → **PIK3R1** targeted by → **SF1126**	49	↓ **slower progression** · Σw -0.090 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.95, n = 37 carriers	**0.57**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

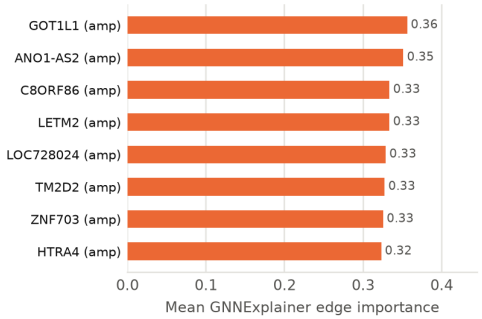
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.125 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	120 pts
+0.097 ↑ risk	NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	50 pts
+0.073 ↑ risk	LowTmb(p) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	73 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.034 ↓ risk	LowAneuploidy(p) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Deactivation of the beta-catenin transactivating complex') ⇒ Progress(p) [KG]	RL-proposed	41 pts
-0.022 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ LinkedTo(g, 'invasive ductal breast carcinoma') ⇒ Progress(p) [KG]	RL-proposed	38 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
CDH1 · driver	Mutation	p.X230_splice	17	0.32
PIK3CA · driver	Mutation	p.Q546R	70	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.36
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.35
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
LETM2	High-level amplification	GISTIC ±2	0	0.33
LOC728024	High-level amplification	GISTIC ±2	0	0.33
TM2D2	High-level amplification	GISTIC ±2	0	0.33
ZNF703	High-level amplification	GISTIC ±2	1	0.33
HTRA4	High-level amplification	GISTIC ±2	0	0.32



47 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 71 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A2DO	54	I	LumB	0.74	no progression, followed 85.3 mo
TCGA-A2-A0YK	61	II	Normal	0.72	no progression, followed 19.3 mo
TCGA-A8-A0A6	64	III	LumA	0.72	no progression, followed 21.0 mo
TCGA-B6-A0IP	74	I	LumA	0.71	no progression, followed 129.1 mo
TCGA-BH-A201	64	I	LumA	0.69	no progression, followed 28.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.